

Fundamental Theorem of Arithmetic in $\mathbb{Q}[x]$

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1. Integers

2. Integers

Lemma 1 (NIBZO). *There is no integer strictly between 0 and 1. Equivalently, if $n \in \mathbb{Z}$ and*

$$0 < n < 1,$$

then such an n cannot exist.

Proof. Suppose, for contradiction, that there is an integer strictly between 0 and 1. Put

$$S = \{n \in \mathbb{Z} : 0 < n < 1\}.$$

Then S is nonempty and $S \subseteq \mathbb{Z}_{\geq 0}$. By the well-ordering principle, S has a least element. Call it m .

Since $0 < m$, multiplicative closure of positive integers gives $0 < m^2$. Since $m < 1$ and $m > 0$, multiplying the inequality $m < 1$ by the positive integer m gives

$$m^2 < m.$$

Thus m^2 is also an integer satisfying

$$0 < m^2 < m < 1.$$

So $m^2 \in S$, but $m^2 < m$. This contradicts the choice of m as the least element of S . Therefore S is empty. $\square$

Definition 1. Define **summation**, or $\sum_{i=i_0}^n a_i$ for $n \in \mathbb{Z}^+$ and $n \geq i_0$ as follows:

$$\sum_{i=i_0}^{i_0} = a_{i_0} \quad \sum_{i=i_0}^n a_i = a_n + \sum_{i=i_0}^{n-1} a_i$$

Definition 2. Define **product**, or $\prod_{i=i_0}^n a_i$ for $n \in \mathbb{Z}^+$ and $n \geq i_0$ as follows:

$$\prod_{i=i_0}^{i_0} = a_{i_0} \quad \prod_{i=i_0}^n a_i = a_n \cdot \prod_{i=i_0}^{n-1} a_i$$

3. Rationals

3.1. Construction

Define $\mathbb{Q} \subseteq \mathbb{Z} \times \mathbb{Z}$, where the elements of $\mathbb{Q}$ are of the form (a, b) where $a, b \in \mathbb{Z}, b \neq 0$. Define two operations $+, \cdot$, such that

- $+: \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$ where $(a, b) + (c, d) = (ad + bc, bd)$
- $\cdot: \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$ where $(a, b) \cdot (c, d) = (ac, bd)$

Also, define an equivalence relation on $\mathbb{Q}$. Two elements $(a, b), (c, d)$ in $\mathbb{Q}$ are equal iff $ad = bc$. Denote equivalence as $(a, b) \sim (c, d)$.

We can also denote (a, b) as $\frac{a}{b}$.

3.2. Ring Axioms

Commutativity of Multiplication:

$$(a, b) \cdot (c, d) = (ac, bd) \quad \text{and} \quad (c, d) \cdot (a, b) = (ca, db) = (ac, bd)$$

Commutativity of Addition:

$$(a, b) + (c, d) = (ad + bc, bd) \quad \text{and} \quad (c, d) + (a, b) = (cb + ad, db) = (ad + bc, bd)$$

Associativity of Addition:

$$((a, b) + (c, d)) + (e, f) = (a, b) + ((c, d) + (e, f))$$

For the left hand side,

$$(ad + bc, bd) + (e, f) = (adf + bcf + bde, bdf)$$

For the right hand side

$$(a, b) + (cf + de, df) = (adf + bcf + bde, bdf).$$

Associativity of Multiplication:

$$(a, b) \cdot ((c, d) \cdot (e, f)) = ((a, b) \cdot (c, d)) \cdot (e, f)$$

For the left hand side,

$$(a, b) \cdot (ce, df) = (ace, bdf)$$

For the right hand side,

$$(ac, bd) \cdot (e, f) = (ace, bdf).$$

Additive Identity:

$$(a, b) + (0, 1) = (a, b)$$

From the definition, $(a, b) + (0, 1)$ equals

$$(a \cdot 1 + b \cdot 0, b \cdot 1)$$

By the one axiom and zero axiom,

$$(a, b)$$

Multiplicative Identity:

$$(a, b) \cdot (1, 1) = (a, b)$$

From the definition,

$$(a, b) \cdot (1, 1)$$

equals

$$(a \cdot 1, b \cdot 1)$$

By the one axiom,

$$(a, b)$$

Distributivity:

We prove for $x, y, z \in \mathbb{Q}$, we have $x(y + z) = xy + xz$. Suppose $x = (a, b), y = (c, d), z = (e, f)$. Then $x(y + z) = (a, b) \cdot ((c, d) + (e, f)) = (a, b) \cdot (cf + de, df) = (a(cf + de), b(df)) = (acf + ade, bdf)$. We also have $xy = (a, b) \cdot (c, d) = (ac, bd)$ and $xz = (a, b) \cdot (e, f) = (ae, bf)$. We then have $xy + xz = (ac, bd) + (ae, bf) = (acbf + bdae, bdbf) = (acf + dae, dbf) = (acf + ade, bdf)$. Hence $x(y + z) = xy + xz$. Right distributivity follows immediately, since $(y + z)x = x(y + z) = xy + xz = yx + zx$.

3.3. Equivalence

$$(a, b) \sim (c, d) \longleftrightarrow ad = bc$$

Symmetry

$$if(a, b) \sim (c, d)$$

then

$$ad = bc$$

and

$$cb = da$$

so

$$(c, d) \sim (a, b)$$

Transitivity

$$(a, b) \sim (c, d), (c, d) \sim (e, f)$$

then

$$ad = bc, cf = de$$

and

$$adc f = bcde$$

and

$$cdf a = cbde$$

so

$$af = be$$

finally

$$(a, b) \sim (e, f)$$

3.4. Well-Defined

To show that addition is well defined, we show that $(a, b) + (c, d) = (a, b) + (e, f)$ if $(c, d) \sim (e, f)$. According to our criterion for an equivalence relation, $cf = de$. Therefore, we want to show $(ad+bc, bd) = (af+be, bf)$. To show these are equivalent, we want to show $abdf+b^2ed = abdf+b^2cf$. We cancel $abdf$ and b^2 because b is nonzero to show that $de = cf$, which is true by our equivalence relation.

To show that multiplication is well defined, we show that $(a, b) \cdot (c, d) = (a, b) \cdot (e, f)$ if $(c, d) \sim (e, f)$. According to our criterion for an equivalence relation, $cf = de$. Therefore, we want to show $(ac, bd) = (ae, bf)$. To show these are equivalent, we want to show $acbf = bdae$. Suppose $a = 0$. Then $0 = 0$, which is true. Otherwise, suppose $a \neq 0$. Then because $ab = ac \implies b = c$ when $a \neq 0$, we can cancel ab from both sides to get $cf = de$, which is true by the equivalence $(c, d) = (e, f)$. Hence multiplication is well defined.

3.5. Multiplicative Inverse

For any (a, b) and $(c, d) \in \mathbb{Q}$, and $a \neq 0$, $\exists(bc, ad)$ where

$$(a, b)(bc, ad) = (abc, abd)$$

and

$$(abc, abd) \sim (c, d)$$

then

$$(a, b) \mid (c, d)$$

Therefore, define $\frac{p}{q}$ for $p, q \in \mathbb{Q}$ and $p \neq 0$ to be the unique solution x to $px = q$. Note also that this is equivalent to the fraction notation of rationals since $\frac{(a,1)}{(b,1)} = (a, b) = \frac{a}{b}$. Then, define p^{-1} to be $\frac{(1,1)}{p}$.

4. The Ring $\mathbb{Q}[x]$

Definition 3. Define the **exponent** $\forall a \in \mathbb{Z}, b \in \mathbb{Z} \cup \{0\}$ as

$$a^b = \begin{cases} 1 & \text{if } b = 0 \\ a \cdot a^{b-1} & \text{otherwise} \end{cases}$$

For properties of exponents, see Appendix B.

Definition 4 (The polynomial ring $\mathbb{Q}[x]$). We define a polynomial $f(x) \in \mathbb{Q}[x]$ with degree n if it can be written as

$$f(x) = \sum_{i=0}^n a_i x^i$$

where $n \in \mathbb{Z}_{\geq 0}$ and each coefficient a_i lies in $\mathbb{Q}$ and $a_n \neq 0$.

4.1. Ring Axioms

Proof. Let

$$f(x) = a_0 + a_1x + \cdots + a_mx^m, \quad g(x) = b_0 + b_1x + \cdots + b_nx^n, \quad h(x) = c_0 + c_1x + \cdots + c_\ell x^\ell.$$

For addition, put missing coefficients equal to 0. Then

$$f(x) + g(x) = (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_N + b_N)x^N$$

for some $N \geq m, n$. Since $\mathbb{Q}$ is closed under addition, each $a_i + b_i$ is rational. Hence $f(x) + g(x) \in \mathbb{Q}[x]$.

For multiplication, for each k with $0 \leq k \leq m + n$, let

$$P_k = \{(i, j) \in \mathbb{Z}^2 : 0 \leq i \leq m, 0 \leq j \leq n, i + j = k\}.$$

Write

$$P_k = \{(i_1, j_1), \dots, (i_t, j_t)\}.$$

Define

$$d_k = a_{i_1} b_{j_1} + \dots + a_{i_t} b_{j_t}.$$

Each d_k is rational, since it is made from finitely many additions and multiplications in $\mathbb{Q}$. Therefore

$$f(x)g(x) = d_0 + d_1x + \dots + d_{m+n}x^{m+n}$$

is in $\mathbb{Q}[x]$.

The zero polynomial is the additive identity, since

$$f(x) + 0_{\mathbb{Q}[x]} = (a_0 + 0) + (a_1 + 0)x + \dots + (a_m + 0)x^m = f(x).$$

The polynomial

$$-f(x) = (-a_0) + (-a_1)x + \dots + (-a_m)x^m$$

is the additive inverse of $f(x)$ because $a_i + (-a_i) = 0$ for every i .

The constant polynomial $1_{\mathbb{Q}[x]}$ is the multiplicative identity. In the product $1_{\mathbb{Q}[x]}f(x)$, the coefficient of x^i is $1 \cdot a_i = a_i$. Hence

$$1_{\mathbb{Q}[x]}f(x) = f(x).$$

Addition is associative and commutative because each coefficient is added in $\mathbb{Q}$, and addition in $\mathbb{Q}$ is associative and commutative.

Multiplication is commutative because the coefficient of x^k in $f(x)g(x)$ comes from all products $a_i b_j$ with $i + j = k$, while the coefficient of x^k in $g(x)f(x)$ comes from all products $b_j a_i$ with $j + i = k$. These are the same products, since multiplication in $\mathbb{Q}$ is commutative.

Multiplication is associative because the coefficient of x^k in $(f(x)g(x))h(x)$ comes from all products

$$(a_i b_j) c_r$$

with $i + j + r = k$, while the coefficient of x^k in $f(x)(g(x)h(x))$ comes from all products

$$a_i (b_j c_r)$$

with $i + j + r = k$. These are equal because multiplication in $\mathbb{Q}$ is associative.

Distributivity holds coefficient by coefficient. Put missing coefficients of $g(x)$ and $h(x)$ equal to 0. Then the coefficient of x^k in $f(x)(g(x) + h(x))$ uses terms of the form

$$a_i(b_j + c_j).$$

Since

$$a_i(b_j + c_j) = a_i b_j + a_i c_j$$

in $\mathbb{Q}$, we get

$$f(x)(g(x) + h(x)) = f(x)g(x) + f(x)h(x).$$

Finally, $0_{\mathbb{Q}[x]}$ has constant coefficient 0, while $1_{\mathbb{Q}[x]}$ has constant coefficient 1. Since $0 \neq 1$ in $\mathbb{Q}$,

$$0_{\mathbb{Q}[x]} \neq 1_{\mathbb{Q}[x]}.$$

□

Definition 5. Define $u(x) \in \mathbb{Q}[x]$ as a **unit** if there exists $v(x) \in \mathbb{Q}[x]$ s.t. $u(x)v(x) = 1$. The set of units in $\mathbb{Q}[x]$ is denoted $\mathbb{Q}[x]^\times$.

Lemma 2. $u(x) \in \mathbb{Q}[x]^\times \Leftrightarrow u(x) \in \mathbb{Q}$

Proof. First, prove $u(x) \in \mathbb{Q} \Rightarrow u(x) \in \mathbb{Q}[x]^\times$. This is a corollary of the Multiplicative Inverse property of $\mathbb{Q}$. Second, prove $u(x) \in \mathbb{Q}[x]^\times \Rightarrow u(x) \in \mathbb{Q}$. Define $u(x) = \sum_{i=0}^n a_i x^i$ and $v(x) = \sum_{j=0}^m b_j x^j$, with n and m the degrees of u and v respectively, then

$$u(x)v(x) = \sum_{i=0}^n \sum_{j=0}^m a_i b_j x^i x^j = \sum_{i=0}^n \sum_{j=0}^m a_i b_j x^{i+j}$$

by Multiplied Sums. Notice that $i + j = m + n$ only once. However, $a_m b_n$ are both nonzero, so $a_m b_n \neq 0$. Thus, we have a nonzero term $a_m b_n x^{m+n} \neq 1$. □

Definition 6. For $f(x) \in \mathbb{Q}[x]/\{0\}$, let the **norm**, or $\|f(x)\|$ be equal to

$$\|f(x)\| = 2^{\deg f(x)}$$

Define $\|0\|$ to be 0. with the following properties:

- $\|f(x)\| \cdot \|g(x)\| = \|f(x)g(x)\|$

Proof. By Multiplied Sums, if $f(x) = \sum_{i=0}^n a_i x^i$ and $g(x) = \sum_{j=0}^m b_j x^j$, with n and m the degrees of f and g respectively, then

$$f(x)g(x) = \sum_{i=0}^n \sum_{j=0}^m a_i b_j x^i x^j = \sum_{i=0}^n \sum_{j=0}^m a_i b_j x^{i+j}$$

Notice that to find the degree of $f(x)g(x)$, we take the maximum value of $i+j$ which is their respective maximums, $n+m$, so:

$$\|f(x)g(x)\| = 2^{n+m} = 2^n 2^m = \|f(x)\| \|g(x)\|$$

□

- For any $c(x) \in \mathbb{Q}[x]^\times$, $\|c(x)\| = 1$.

Proof. Since $\deg c(x) = 0$, $\|c(x)\| = 2^0 = 1$.

□

- $\|f(x)\|$ is always positive or 0. Since $2 \in \mathbb{Z}^+$, 2^n for $n \in \mathbb{Z}^+$ is always positive by Appendix B, the norm $\in \mathbb{Z} \cup \{0\}$

Definition 7 (Zero, one, and negatives). The zero element of $\mathbb{Q}[x]$ is the polynomial $0_{\mathbb{Q}[x]} = 0$. Define the degree of $0_{\mathbb{Q}[x]}$ to be $-\infty$. The one element of $\mathbb{Q}[x]$ is the constant polynomial

$$1_{\mathbb{Q}[x]} = 1$$

If

$$f(x) = \sum_{i=0}^n a_i x^i$$

then

$$-f(x) = \sum_{i=0}^n -a_i x^i$$

Theorem 3 ($\mathbb{Q}[x]$ has no zero divisors). If $f(x), g(x) \in \mathbb{Q}[x]$ and

$$f(x)g(x) = 0,$$

then $f(x) = 0$ or $g(x) = 0$.

Proof. Suppose, for contradiction, that

$$f(x)g(x) = 0$$

with $f(x) \neq 0$ and $g(x) \neq 0$.

Since $g(x) \neq 0$, by the norm lemma,

$$\|f(x)\| \leq \|f(x)g(x)\|.$$

Since $f(x) \neq 0$, we have

$$\|f(x)\| = 2^{\deg(f(x))} > 0.$$

So $\|f(x)g(x)\|$ is positive. In particular, $f(x)g(x) \neq 0$, since the norm is only positive for nonzero polynomials.

This contradicts $f(x)g(x) = 0$. Therefore at least one of $f(x)$ and $g(x)$ must be zero. $\square$

Corollary 4 (Cancellation). *If $h(x) \neq 0$ and*

$$h(x)f(x) = h(x)g(x),$$

then $f(x) = g(x)$.

Proof. Subtract the right side from the left:

$$h(f - g) = 0.$$

Since $h \neq 0$ and $\mathbb{Q}[x]$ has no zero divisors, we must have $f - g = 0$. Thus $f = g$. $\square$

4.2. Basic Arithmetic

Lemma 5 (Uniqueness of zero and one). *The additive identity and multiplicative identity in $\mathbb{Q}[x]$ are unique. In particular, if $f(x) + z(x) = f(x)$ for every $f(x) \in \mathbb{Q}[x]$, then $z(x) = 0$. Similarly, if $f(x)u(x) = f(x)$ for every $f(x) \in \mathbb{Q}[x]$, then $u(x) = 1$.*

Proof. Suppose z acts like an additive identity for f , so $f + z = f$. Add $-f$ to both sides:

$$(-f) + (f + z) = (-f) + f.$$

By associativity and the definition of additive inverse, this becomes

$$0 + z = 0.$$

Since 0 is the additive identity, $z = 0$.

Now suppose u acts like a multiplicative identity for every polynomial. Test this on the constant polynomial 1:

$$1 \cdot u = 1.$$

But $1 \cdot u = u$. Hence $u = 1$. □

Lemma 6 (Uniqueness of additive inverses). *If $u(x)$ and $v(x)$ are both additive inverses of $f(x)$, then $u(x) = v(x)$.*

Proof. Assume

$$f + u = 0 \quad \text{and} \quad f + v = 0.$$

Then $f + u = f + v$. Add $-f$ to both sides:

$$(-f) + (f + u) = (-f) + (f + v).$$

By associativity,

$$((-f) + f) + u = ((-f) + f) + v.$$

Thus

$$0 + u = 0 + v,$$

so $u = v$. □

Lemma 7 (Zero and negatives). *For all $f(x), g(x) \in \mathbb{Q}[x]$,*

$$f(x) \cdot 0 = 0, \quad -(-f(x)) = f(x), \quad (-f(x))(-g(x)) = f(x)g(x).$$

Proof. First,

$$f \cdot 0 = f(0 + 0) = f \cdot 0 + f \cdot 0.$$

Adding the additive inverse of $f \cdot 0$ to both sides gives $f \cdot 0 = 0$.

Next, since

$$(-f) + f = 0,$$

the polynomial f is an additive inverse of $-f$. By uniqueness of additive inverses,

$$-(-f) = f.$$

For the product of negatives, first observe that

$$fg + (-f)g = (f + (-f))g = 0 \cdot g = 0.$$

Therefore $(-f)g = -(fg)$. Applying this with g replaced by $-g$ gives

$$(-f)(-g) = -(f(-g)).$$

Also,

$$fg + f(-g) = f(g + (-g)) = f \cdot 0 = 0,$$

so $f(-g) = -(fg)$. Hence

$$(-f)(-g) = -(-(fg)) = fg.$$

□

Lemma 8. *Let $f(x), g(x) \in \mathbb{Q}[x]$ with $f(x) \neq 0$ and $g(x) \neq 0$. Then*

$$\|f(x)\| \leq \|f(x)g(x)\|.$$

Proof. Let

$$\deg(f) = m \quad \text{and} \quad \deg(g) = n.$$

Since $g(x) \neq 0$, we have $n \geq 0$. Also, since $f(x) \neq 0$ and $g(x) \neq 0$, the leading coefficient of $f(x)g(x)$ is the product of the leading coefficients of $f(x)$ and $g(x)$, which is nonzero in $\mathbb{Q}$. Thus

$$\deg(fg) = \deg(f) + \deg(g) = m + n.$$

Therefore

$$\|f(x)g(x)\| = 2^{\deg(fg)} = 2^{m+n}.$$

Since $n \geq 0$,

$$2^m \leq 2^{m+n}.$$

So

$$\|f(x)\| = 2^m \leq 2^{m+n} = \|f(x)g(x)\|.$$

□

4.3. Divisibility

Definition 8 (Divisibility). For $f(x), g(x) \in \mathbb{Q}[x]$, we say $f(x)$ divides $g(x)$, written

$$f(x) \mid g(x),$$

if there exists $q(x) \in \mathbb{Q}[x]$ such that

$$g(x) = f(x)q(x).$$

Definition 9 (Greatest Common Divisor). For $f(x), g(x) \in \mathbb{Q}[x]$, we define $\gcd(f(x), g(x)) = d(x)$ to satisfy:

- $d(x) \mid f(x)$ and $d(x) \mid g(x)$
- $\forall h(x) \in \mathbb{Q}[x]$ s.t. $h(x) \mid f(x)$ and $h(x) \mid g(x)$, $\|h(x)\| \leq \|d(x)\|$

Theorem 9 (Existence and uniqueness of gcd). *Let $f(x), g(x) \in \mathbb{Q}[x]$, not both zero. Then there exists a unique monic polynomial $d(x) \in \mathbb{Q}[x]$ such that*

$$d(x) = \gcd(f(x), g(x)).$$

Proof. Let

$$S = \{ a(x)f(x) + b(x)g(x) : a(x), b(x) \in \mathbb{Q}[x], a(x)f(x) + b(x)g(x) \neq 0 \}.$$

Since $f(x)$ and $g(x)$ are not both zero, the set S is nonempty. Also, every element of S has positive norm. So the set

$$N = \{ \|s(x)\| : s(x) \in S \}$$

is a nonempty subset of $\mathbb{Z}^+$. By the well-ordering principle, N has a least element. Choose $d(x) \in S$ with smallest norm. Then for some $a(x), b(x) \in \mathbb{Q}[x]$,

$$d(x) = a(x)f(x) + b(x)g(x).$$

We first show that $d(x) \mid f(x)$. By the division algorithm, there exist $q(x), r(x) \in \mathbb{Q}[x]$ such that

$$f(x) = q(x)d(x) + r(x),$$

where either $r(x) = 0$ or

$$\|r(x)\| < \|d(x)\|.$$

Rearranging gives

$$r(x) = f(x) - q(x)d(x).$$

Since $d(x) = a(x)f(x) + b(x)g(x)$, we get

$$r(x) = f(x) - q(x)(a(x)f(x) + b(x)g(x)).$$

Thus

$$r(x) = (1 - q(x)a(x))f(x) + (-q(x)b(x))g(x).$$

So if $r(x) \neq 0$, then $r(x) \in S$, but it has smaller norm than $d(x)$, which contradicts how $d(x)$ was chosen. Therefore $r(x) = 0$, and hence

$$d(x) \mid f(x).$$

The same argument with $g(x)$ in place of $f(x)$ shows that

$$d(x) \mid g(x).$$

Now let $h(x) \in \mathbb{Q}[x]$ be any common divisor of $f(x)$ and $g(x)$. Then

$$f(x) = h(x)u(x) \quad \text{and} \quad g(x) = h(x)v(x)$$

for some $u(x), v(x) \in \mathbb{Q}[x]$. Hence

$$d(x) = a(x)f(x) + b(x)g(x) = a(x)h(x)u(x) + b(x)h(x)v(x).$$

Factoring gives

$$d(x) = h(x)(a(x)u(x) + b(x)v(x)).$$

Thus

$$h(x) \mid d(x).$$

So $d(x)$ satisfies the gcd property.

Finally, make $d(x)$ monic. Let $c \in \mathbb{Q}$ be the leading coefficient of $d(x)$. Since $d(x) \neq 0$, we have $c \neq 0$. Define

$$d_0(x) = c^{-1}d(x).$$

Then $d_0(x)$ is monic. Since c is a nonzero rational number, multiplying by c or c^{-1} does not change divisibility in $\mathbb{Q}[x]$. Therefore $d_0(x)$ is also a greatest common divisor of $f(x)$ and $g(x)$.

It remains to prove uniqueness. Suppose $d_1(x)$ and $d_2(x)$ are both monic greatest common divisors of $f(x)$ and $g(x)$. Since $d_1(x)$ is a gcd and $d_2(x)$ is a common divisor, we have

$$d_2(x) \mid d_1(x).$$

Similarly,

$$d_1(x) \mid d_2(x).$$

So there exist $u(x), v(x) \in \mathbb{Q}[x]$ such that

$$d_1(x) = d_2(x)u(x) \quad \text{and} \quad d_2(x) = d_1(x)v(x).$$

Taking degrees gives

$$\deg(d_1) = \deg(d_2) + \deg(u)$$

and

$$\deg(d_2) = \deg(d_1) + \deg(v).$$

Thus $\deg(u) = \deg(v) = 0$, so $u(x)$ and $v(x)$ are nonzero constant polynomials. Since $d_1(x)$ and $d_2(x)$ are both monic, the constant $u(x)$ must be 1. Therefore

$$d_1(x) = d_2(x).$$

Hence the monic greatest common divisor exists and is unique. $\square$

5. The Division Algorithm in $\mathbb{Q}[x]$

For $\mathbb{Q}[x]$, the norm plays the role that size plays in integer division. Since norms of nonzero polynomials are positive integers, we can use the well-ordering principle to pick a remainder of smallest norm.

Theorem 10 (Division algorithm in $\mathbb{Q}[x]$). *Let $f(x), g(x) \in \mathbb{Q}[x]$ with $g(x) \neq 0$. Then there exist unique polynomials $q(x), r(x) \in \mathbb{Q}[x]$ such that*

$$f(x) = q(x)g(x) + r(x),$$

where either

$$r(x) = 0$$

or

$$\|r(x)\| < \|g(x)\|.$$

Proof. Existence. Look at all possible remainders after subtracting multiples of g from f :

$$S = \{ f(x) - q(x)g(x) : q(x) \in \mathbb{Q}[x] \}.$$

If $0 \in S$, then $f = qg$ for some q , so

$$f = qg + 0,$$

and the result holds with $r = 0$.

Now suppose $0 \notin S$. Then every element of S is nonzero, so each element has a norm in $\mathbb{Z}^+$. Define

$$D = \{ \|s(x)\| : s(x) \in S \} \subseteq \mathbb{Z}^+.$$

The set D is nonempty because $f - 0g = f$ lies in S . By the well-ordering principle, D has a

smallest element. Choose $r \in S$ with minimum norm. Since $r \in S$, there exists $q \in \mathbb{Q}[x]$ such that

$$r = f - qg.$$

Equivalently,

$$f = qg + r.$$

It remains to show that $\|r(x)\| < \|g(x)\|$. Suppose instead that

$$\|r(x)\| \geq \|g(x)\|.$$

Write

$$r(x) = a_m x^m + \cdots \quad \text{and} \quad g(x) = b_n x^n + \cdots,$$

where $a_m, b_n \neq 0$, $m = \deg(r)$, and $n = \deg(g)$. Since $\|r(x)\| \geq \|g(x)\|$ means $2^m \geq 2^n$, we get $m \geq n$. Since $b_n \neq 0$ and $b_n \in \mathbb{Q}$, the rational number a_m/b_n exists. Define

$$r^*(x) = r(x) - \frac{a_m}{b_n} x^{m-n} g(x).$$

The second term has leading term $a_m x^m$, so the leading term of r cancels. Thus either $r^* = 0$ or

$$\|r^*(x)\| < \|r(x)\|.$$

Also,

$$\begin{aligned} r^* &= f - qg - \frac{a_m}{b_n} x^{m-n} g \\ &= f - \left(q + \frac{a_m}{b_n} x^{m-n} \right) g. \end{aligned}$$

Therefore $r^* \in S$. If $r^* = 0$, then $0 \in S$, which is not the case we are in. If $r^* \neq 0$, then r^* is in S and has smaller norm than r , contradicting how we chose r . Therefore our assumption was false, and

$$\|r(x)\| < \|g(x)\|.$$

Uniqueness. Suppose

$$f = q_1 g + r_1 = q_2 g + r_2,$$

where r_1 and r_2 are both either zero or have norm less than $\|g(x)\|$. Subtracting gives

$$g(q_1 - q_2) = r_2 - r_1.$$

Assume $q_1 \neq q_2$. Then $q_1 - q_2 \neq 0$, and since $\mathbb{Q}[x]$ has no zero divisors,

$$g(q_1 - q_2) \neq 0.$$

Hence

$$\begin{aligned} \|g(x)(q_1(x) - q_2(x))\| &= 2^{\deg(g(q_1 - q_2))} \\ &= 2^{\deg(g) + \deg(q_1 - q_2)} \\ &= \|g(x)\| \cdot \|q_1(x) - q_2(x)\| \\ &\geq \|g(x)\|. \end{aligned}$$

But $r_2 - r_1$ is either zero or has norm strictly less than $\|g(x)\|$, since each of r_1 and r_2 is either zero or has norm less than $\|g(x)\|$, so their degrees are both less than $\deg(g)$ when they are nonzero. This is impossible. Therefore $q_1 = q_2$. Substituting back gives $r_1 = r_2$. $\square$

6. Euclid's Algorithm

Definition 10. Define **Euclids Algorithm** on $f(x), g(x) \in \mathbb{Q}[x]$ to produce $r, q \in \mathbb{Q}[x]$ such that

$$f = q_1g + r_1 \quad g = q_2r_1 + r_2 \quad r_{n-2} = q_nr_{n-1} + r_n$$

Lemma 11. For any $f(x), g(x) \in \mathbb{Q}[x]$, the Euclidean Algorithm generates an $n \in \mathbb{Z}^+$ such that $r_n = 0$.

Proof. Let $S = \{f(x) \in \mathbb{Q}[x] \mid \exists g(x) \text{ s.t. } r_i \neq 0 \forall i \in \mathbb{Z}\}$. For the sake of contradiction, S is not empty. By the well-ordering principle, there exists a least element l . Performing Euclid's Algorithm on (l, g) , we have:

$$\begin{aligned} g &= q_1l + r_1 \\ l &= q_2r_1 + r_2 \end{aligned}$$

If (g, l) does not reach 0, (l, r_1) will also not reach 0, as they generate the same values. Thus, $r_1 \in S$, but $r_1 < l$ by the division algorithm, which is a contradiction. Thus, S must be empty. $\square$

Lemma 12. For any $f(x), g(x) \in \mathbb{Q}[x]$, if the Euclidean Algorithm generates $r_n = 0$, $r_{n-1} = \gcd(f, g)$.

Proof. Let $S = \{f(x) \in \mathbb{Q}[x] \mid \exists g(x) \text{ s.t. } r_{n-1} \neq \gcd(f, g)\}$. For the sake of contradiction, S is not empty. By the well-ordering principle, there exists a least element l . Performing Euclid's Algorithm

on (l, g) , we have:

$$g = q_1 l + r_1$$

If $\gcd(g, l) = d$, $d \mid g$ and $d \mid l$. Since $b > l$, $\gcd(l, r_1)$ remains d . However, the Euclidean Algorithm on (r_1, l) generates the same values as (b, l) . Thus, $r_1 \in S$, but $r_1 < l$ by the division algorithm, which is a contradiction. Thus, S must be empty. $\square$

Theorem 13. *Each remainder in Euclid's algorithm on $f(x), g(x) \in \mathbb{Q}[x]$ can be expressed as a linear combination of $f(x)$ and $g(x)$, or*

$$a(x), b(x) \in \mathbb{Q}[x] \quad f(x)a_n(x) + g(x)b_n(x) = r_n(x)$$

Proof. Since $f = q_1 g + r_1$, $r_1 = f - q_1 g$. In addition, since $r_2 = g - q_2 r_1 = g - q_2 g + q_1 q_2 f$, r_2 can also be expressed as a linear combination of $f(x)$ and $g(x)$. Let

$$S := \{k \in \mathbb{Z}^+ \mid r_k \neq f(x)a(x) + g(x)b(x) \forall a(x), b(x) \in \mathbb{Q}[x]\}$$

For the sake of contradiction, S is not empty. Thus, by the well-ordering principle, there exists a least element $l \in S$. However, $1, 2 \notin S$, so $l > 2$ and $l - 2, l - 1 \in \mathbb{Z}^+$. Since l is the least element of S , $l - 2, l - 1 \notin S$.

$$\begin{aligned} \exists a_1(x), b_1(x) \in \mathbb{Q}[x] \quad a_1 f + b_1 g &= r_{l-1} \\ \exists a_2(x), b_2(x) \in \mathbb{Q}[x] \quad a_2 f + b_2 g &= r_{l-2} \end{aligned}$$

Thus, by the Euclidean Algorithm, we know that

$$\begin{aligned} r_{l-2} &= q_l r_{l-1} + r_l \\ a_2 f + b_2 g &= q_l (a_1 f + b_1 g) + r_l \\ r_l &= (a_2 - q_l a_1) f + (b_2 - q_l b_1) g \end{aligned}$$

As $(a_2 - q_l a_1), (b_2 - q_l b_1) \in \mathbb{Q}[x]$, r_l can thus be written as a linear combination of $f(x)$ and $g(x)$. However, since $l \in S$, this is a contradiction and S must be empty. $\square$

Corollary 14. *(Bezout's) The greatest common divisor of $f(x), g(x) \in \mathbb{Q}[x]$ can be expressed as a*

linear combination of $f(x)$ and $g(x)$, or

$$k_1(x), k_2(x) \in \mathbb{Q}[x] \quad f(x)k_1(x) + g(x)k_2(x) = \gcd(f(x), g(x))$$

Proof. With $r_n = 0$, the Euclidean algorithm always produces $r_{n-1} = \gcd(f(x), g(x))$. $\square$

7. Fundamental Lemma

7.1. Definition

Define $p(x) \in \mathbb{Q}[x]$ to be prime if:

- $\forall a(x), b(x) \in \mathbb{Q}[x]$ where $a(x)b(x) = p(x)$, either $a(x)$ or $b(x)$ is a unit
- $p(x)$ is not a unit

Note that all linears $l(x) \in \mathbb{Q}[x]$ (polynomials of degree 1) are prime in $\mathbb{Q}[x]$ since they have a norm of 2. for any $a(x), b(x) \in \mathbb{Q}[x]$ where $a(x)b(x) = l(x)$, $\|a(x)\|\|b(x)\| = 2$ (norm preserves multiplication), and $\|a(x)\|, \|b(x)\| \in \mathbb{Z}^+$ (since they cannot be 0). Then, because 2 is prime in $\mathbb{Z}^+$, $\|a(x)\| = 1 \vee \|b(x)\| = 1$. This means that either a or b is a unit, and it follows then that $l(x)$ is a prime.

7.2. Prime Factors

Theorem 15 (Existence of prime factors). $\forall a(x) \in \mathbb{Q}[x]$ where $a(x)$ isn't a unit, $\exists p(x)$ which is prime where $p(x) | a(x)$.

When $a(x) = 0$, it is divisible by all primes. For some polynomial $a(x)$ which isn't 0 or a unit, let $S = \{n(x) \in \mathbb{Q}[x] \mid n(x) \neq 0 \wedge \|n(x)\| \neq 1 \wedge n(x) \mid a(x)\}$. We know S is non-empty since $a(x) \in S$.

Then consider $\|S\| = \{\|n(x)\| \in S\}$. Since $\forall n(x) \in \mathbb{Q}[x]$ where $n \neq 0$, $\|n(x)\| \in \mathbb{Z}^+$, $\|S\| \in \mathbb{Z}^+$. Then, since it is non-empty, there must exist an element $\|m(x)\| \in \|S\|$ which corresponds to $m(x) \in S$ where $\|m(x)\|$ is minimal in $\|S\|$.

Suppose $\exists k(x) \mid m(x)$, where $\|k(x)\| \neq \|m(x)\|$ and isn't a unit. Since $\|k(x)\| \mid \|m(x)\|$ and $\|k(x)\|, \|m(x)\| \in \mathbb{Z}^+$, $\|k(x)\| \leq \|m(x)\|$, but $\|k(x)\| \neq \|m(x)\|$, so $\|k(x)\| < \|m(x)\|$. Since $k(x) \mid m(x)$, then because $m(x) \mid a(x)$, $k(x) \mid a(x)$, so $k(x) \in S$. Then, we have $\|k(x)\| \in \|S\|$ too, but $\|k(x)\| < \|m(x)\|$, so $\|m(x)\|$ is not minimal in $S \rightarrow \leftarrow$.

Therefore, if $\exists k(x) \mid m(x)$, $\|k(x)\| = \|m(x)\|$ or it is a unit. For any $a(x)b(x) = m(x)$, either $a(x)$ is a unit or $\|a(x)\| = \|m(x)\|$. In the latter case, this implies that $b(x)$ is a unit, since preservation of norm after multiplication is unique to units. Consequently, $\forall a(x), b(x) \in \mathbb{Q}[x]$ where $a(x)b(x) = m(x)$, either $a(x)$ or $b(x)$ is a unit, and $m(x)$ isn't a unit, meaning that $m(x)$ is prime.

7.3. Prime Factorisation

Theorem 16 (Existence of a Prime Factorisation). $\forall n(x) \in \mathbb{Q}[x], \exists S \subset \mathbb{Q}[x]$ where $\forall a(x) \in S$, $a(x)$ is prime, such that $n(x) = \prod_{a(x) \in S} a(x)$.

$\forall n(x) \in \mathbb{Z}^+$, $\exists m_1(x) \mid n(x)$ and $m_1(x)$ is prime. If $m_1(x) = n(x)$, then such S exists where $S = \{m_1\}$. Otherwise, $\exists n_1(x)$ where $m_1(x)n_1(x) = n(x)$. It follows then that $\|m_1(x)\| \|n_1(x)\| = \|n(x)\|$, so $\|n_1(x)\| \mid \|n(x)\|$. Since they are both positive integers, this means that $\|n_1(x)\| \leq \|n(x)\|$. However, if $\|n_1(x)\| = \|n(x)\|$, then $\|m_1(x)\| = 1$, meaning it is a unit and therefore can't be prime. Therefore, $\|n_1(x)\| < \|n(x)\|$.

Similarly, either $n_1(x)$ is prime or $m_2(x)n_2(x) = n_1(x)$, where $m_2(x)$ is prime. If this terminates at $n_k(x)$ which is prime, we have $n(x) = n_k(x) \prod_{i=1}^k m_i(x)$, and all terms on the right hand side are prime. Then, we can have $S = \{n_k(x)\} \cup \{m_i \mid i \in \mathbb{Z} \wedge 0 < i \leq k\}$.

Suppose this doesn't terminate. Then, we have set $\|N\| = \{\|n_i(x)\| \mid i \in \mathbb{Z}^+\}$ which is infinite, and $\forall k \in \mathbb{Z}^+$, $\|n_k(x)\| > \|n_{k+1}(x)\|$. Also, $\|N\| \subset \mathbb{Z}^+$ since the norm of all non-zero polynomials is a positive integer. Since it is non-empty, there must be a minimal element $\|n_m(x)\|$ in it. However, $\|n_m(x)\| > \|n_{m+1}(x)\|$, so it isn't minimal $\rightarrow \leftarrow$.

Therefore, it must terminate, in which case there exists such set S .

Lemma 17 (Euclid's Lemma in $\mathbb{Q}[x]$). Let $p(x), f(x), g(x) \in \mathbb{Q}[x]$, and suppose $p(x)$ is irreducible. If $p(x) \mid f(x)g(x)$, then $p(x) \mid f(x)$ or $p(x) \mid g(x)$.

Proof. Suppose $p(x) \mid f(x)g(x)$. If $p(x) \mid f(x)$, then we are done. So assume $p(x) \nmid f(x)$.

Let $d(x) = \gcd(p(x), f(x))$. Since $d(x) \mid p(x)$ and $p(x)$ is irreducible, $d(x)$ is either a unit or an associate of $p(x)$.

If $d(x)$ were an associate of $p(x)$, then $p(x) \mid d(x)$. Since $d(x) \mid f(x)$, this would imply $p(x) \mid f(x)$, contradicting our assumption. Therefore $d(x)$ is a unit.

Since gcds are chosen to be monic, the only monic unit is 1. Hence $d(x) = 1$. By Bezout's

Identity, there exist $a(x), b(x) \in \mathbb{Q}[x]$ such that

$$a(x)p(x) + b(x)f(x) = 1.$$

Multiplying both sides by $g(x)$ gives

$$a(x)p(x)g(x) + b(x)f(x)g(x) = g(x).$$

Now $p(x)$ divides $a(x)p(x)g(x)$ because that term has a factor of $p(x)$. Also, $p(x)$ divides $b(x)f(x)g(x)$ because $p(x) \mid f(x)g(x)$. Therefore $p(x)$ divides their sum, so $p(x) \mid g(x)$.

Thus $p(x) \mid f(x)$ or $p(x) \mid g(x)$. □

8. Conclusion

Conclude with the big ideas!

Appendix A: Summation and Product Lemmas

Lemma 18. $\sum_{i=i_0}^n a_i + \sum_{i=i_0}^n b_i = \sum_{i=i_0}^n (a_i + b_i)$

Proof. Note that

$$\sum_{i=i_0}^{i_0} a_i + \sum_{i=i_0}^{i_0} b_i = a_{i_0} + b_{i_0} = \sum_{i=i_0}^{i_0} (a_i + b_i)$$

Let $S = \left\{ k \in \mathbb{Z}^+ \mid \sum_{i=i_0}^{i_0+k} a_i + \sum_{i=i_0}^{i_0+k} b_i \neq \sum_{i=i_0}^{i_0+k} (a_i + b_i) \right\}$. For the sake of contradiction, S is not empty.

As $S \subseteq \mathbb{Z}^+$, by the Well-Ordering Principle, there exists a least element $l \in S$.

If $k = 1$:

$$\sum_{i=i_0}^{i_0+1} a_i + \sum_{i=i_0}^{i_0+1} b_i = a_{i_0+1} + a_{i_0} + b_{i_0+1} + b_{i_0} = (a_{i_0+1} + b_{i_0+1}) + (a_{i_0} + b_{i_0}) = \sum_{i=i_0}^{i_0+1} (a_i + b_i)$$

Thus, $1 \notin S$ and so $l > 1$ by NIBZO. Since $l - 1 \in \mathbb{Z}$ yet $l - 1 \notin S$,

$$\begin{aligned} \sum_{i=i_0}^{i_0+l-1} a_i + \sum_{i=i_0}^{i_0+l-1} b_i &= \sum_{i=i_0}^{i_0+l-1} (a_i + b_i) \\ a_l + b_l + \sum_{i=i_0}^{i_0+l-1} a_i + \sum_{i=i_0}^{i_0+l-1} b_i &= a_l + b_l + \sum_{i=i_0}^{i_0+l-1} (a_i + b_i) \\ a_l + \sum_{i=i_0}^{i_0+l-1} a_i + b_l + \sum_{i=i_0}^{i_0+l-1} b_i &= (a_l + b_l) + \sum_{i=i_0}^{i_0+l-1} (a_i + b_i) \\ \sum_{i=i_0}^{i_0+l} a_i + \sum_{i=i_0}^{i_0+l} b_i &= \sum_{i=i_0}^{i_0+l} (a_i + b_i) \end{aligned}$$

However, since $l \in S$, this is a contradiction. Thus, S must be empty. $\square$

Lemma 19. *Sum Constants:* $\sum_{i=i_0}^n ca_i = c \sum_{i=i_0}^n a_i$

Proof. Note that

$$\sum_{i=i_0}^{i_0} ca_i = ca_{i_0} = c \sum_{i=i_0}^{i_0} a_i$$

and that

$$\sum_{i=i_0}^{i_0+1} ca_i = ca_{i_0} + ca_{i_0+1} = c \sum_{i=i_0}^{i_0+1} a_i$$

Let $S = \left\{ k \in \mathbb{Z}^+ \mid \sum_{i=i_0}^{i_0+k} ca_i \neq c \sum_{i=i_0}^{i_0+k} a_i \right\}$. For the sake of contradiction, S is not empty. As $S \subseteq \mathbb{Z}^+$, by the well-ordering principle, there exists a least element $l \in S$. Since $1 \notin S$, $l > 1$ by NIBZO. As $l - 1 \in \mathbb{Z}$ yet $l - 1 \notin S$,

$$\begin{aligned} \sum_{i=i_0}^{i_0+l-1} ca_i &= c \sum_{i=i_0}^{i_0+l-1} a_i \\ ca_l + \sum_{i=i_0}^{i_0+l-1} ca_i &= ca_l + c \sum_{i=i_0}^{i_0+l-1} a_i \\ \sum_{i=i_0}^{i_0+l} ca_i &= c \sum_{i=i_0}^{i_0+l} a_i \end{aligned}$$

However, since $l \in S$, this is a contradiction. Thus, S must be empty. $\square$

Lemma 20. *Multiplied Sums:* $\sum_{i=i_0}^n a_i \cdot \sum_{j=j_0}^m b_j = \sum_{i=i_0}^n \sum_{j=j_0}^m (a_i b_j)$

Proof. By Sum Constants, we know that

$$\begin{aligned} \sum_{i=i_0}^n a_i \cdot \sum_{j=j_0}^m b_j &= \sum_{i=i_0}^n (a_i \cdot \sum_{j=j_0}^m b_j) \\ \sum_{i=i_0}^n a_i \cdot \sum_{j=j_0}^m b_j &= \sum_{i=i_0}^n (\sum_{j=j_0}^m a_i b_j) \end{aligned}$$

□

Lemma 21. $\prod_{i=i_0}^n a_i \prod_{i=i_0}^n b_i = \prod_{i=i_0}^n a_i b_i$

Proof. Note that

$$\prod_{i=i_0}^{i_0} a_i \prod_{i=i_0}^{i_0} b_i = a_{i_0} b_{i_0} = \prod_{i=i_0}^{i_0} a_i b_i$$

Let $S = \left\{ k \in \mathbb{Z}^+ \mid \prod_{i=i_0}^{i_0+k} a_i \prod_{i=i_0}^{i_0+k} b_i \neq \prod_{i=i_0}^{i_0+k} a_i b_i \right\}$. For the sake of contradiction, S is not empty. As $S \subseteq \mathbb{Z}^+$, by the Well-Ordering Principle, there exists a least element $l \in S$.

If $k = 1$:

$$\prod_{i=i_0}^{i_0+1} a_i \prod_{i=i_0}^{i_0+1} b_i = a_{i_0+1} a_{i_0} \cdot b_{i_0+1} b_{i_0} = (a_{i_0+1} b_{i_0+1})(a_{i_0} b_{i_0}) = \prod_{i=i_0}^{i_0+1} a_i b_i$$

Thus, $1 \notin S$ and so $l > 1$ by NIBZO. Since $l - 1 \in \mathbb{Z}$ yet $l - 1 \notin S$,

$$\begin{aligned} \prod_{i=i_0}^{i_0+l-1} a_i \prod_{i=i_0}^{i_0+l-1} b_i &= \prod_{i=i_0}^{i_0+l-1} a_i b_i \\ a_l b_l \cdot \prod_{i=i_0}^{i_0+l-1} a_i \prod_{i=i_0}^{i_0+l-1} b_i &= a_l b_l \cdot \prod_{i=i_0}^{i_0+l-1} a_i b_i \\ (a_l \prod_{i=i_0}^{i_0+l-1} a_i) \cdot (b_l \prod_{i=i_0}^{i_0+l-1} b_i) &= (a_l b_l) \prod_{i=i_0}^{i_0+l-1} a_i b_i \\ \prod_{i=i_0}^{i_0+l} a_i \prod_{i=i_0}^{i_0+l} b_i &= \prod_{i=i_0}^{i_0+l} a_i b_i \end{aligned}$$

However, since $l \in S$, this is a contradiction. Thus, S must be empty. □

Appendix B: Exponent Lemmas

Lemma 22. $a^n a^m = a^{n+m} \quad \forall a \in \mathbb{Z}, n, m \in \mathbb{Z} \cup \{0\}$

Proof. Let $S = \{n \in \mathbb{Z}^+ \mid \exists m \in \mathbb{Z}^+ \cup \{0\} \text{ s.t. } a^n a^m \neq a^{n+m}\}$. For the sake of contradiction, S is not empty. By the well-ordering principle, since $S \subset \mathbb{Z}^+$, there must be a least element $l \in S$. First, $\forall m \in \mathbb{Z}^+$,

$$a \cdot a^{m-1} = a^m$$

$$a^1 \cdot a^{m-1} = a^m$$

Thus, $1 \notin S$ as $m - 1 \in \mathbb{Z}^+$. Therefore, by NIBZO $l > 1$ and $(l - 1) \in \mathbb{Z}^+$, but $l - 1 \notin S$.

$$a^{l-1} a^m = a^{l+m-1}$$

$$a \cdot a^{l-1} a^m = a \cdot a^{l+m-1}$$

$$a^l a^m = a^{l+m}$$

Thus, $l \notin S$. However, this contradicts that l is the least element of S . So, S must be empty. $\square$